STOCHASTIC METHODS IN WATER RESOURCES

Unit 3: Random functions

Lecture 9: Definitions, random function types, stationary random functions

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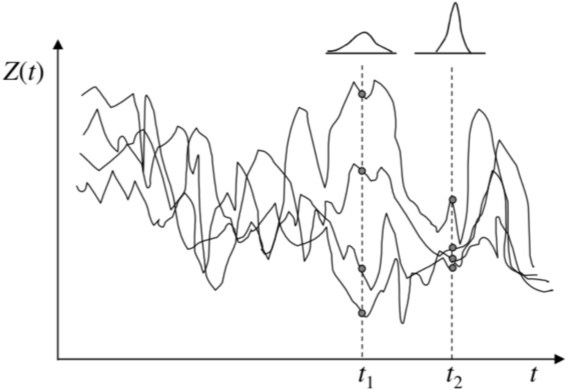
Definitions

- Consider a hydrological variable Z that can either varies in space (e.g. hydraulic conductivity) or in time (e.g. surface water level) or in space and time (e.g. growndwater depth).
- The value of any of these variables is known only in those places where it is observed, and it is unknown in other places.
- Accordingly, the value of Z is uncertain, to deal with this, instead of using one single functio either in time ($Z(t)$), or space ($Z(\mathbf{x})$) or space and time ($Z(\mathbf{x}, t)$), multiple functions are used (see figure for $Z(t)$).
- Each of these functions have the same probability of representing the true.
- The family of equally probable functions representing Z is known as an **emsemble** or a **random function**.



Definitions

- **Random field**: If Z is a function of space ($\mathbf{x}$), $Z(\mathbf{x})$, $\mathbf{x} = (x, y, z)$.
- **Random or stochastic process**: If Z is a function of time (t), $Z(t)$.
- **Space-time random field**: If Z is a function of space ($\mathbf{x}$) and time (t), $Z(\mathbf{x}, t)$, $\mathbf{x} = (x, y, z)$.

Here, we are going to focus on random process ($Z(t)$). Note that in the figure above, four functions $Z(t)$ (ensemble members) are shown and each of them are know as realizations and denoted as $z(t)$. The number of realizations, depending on the variable's nature, can be finite or infinite.
Another way of explaining this concept is to imagine a random function as a collection of random variables (one at every location in space or point in time). Observing at the figure above, for each time t, a $Z(t)$ is defined. This is why for each t, $Z(t)$ is described with a pdf $f_Z(z;t)$, which depends not only on the value of z but also on the value of t. These pdf is estimated based on sample of data taken at particular time (e.g. t_1). As shown in the figure, the pdfs of the data sample at t_1 and t_2 are different, and indicate that variance is larger in t_1 than in t_2 and thus more uncertain.
At each point in time (t), μ_Z is calculated as:

$$\mu_Z(t) = E[Z(t)] = \int_{-\infty}^{\infty} z f_Z(z;t) dz$$

and σ_Z^2 is:

$$\sigma_Z^2(t) = E[(Z(t) - \mu_Z(t))^2] = \int_{-\infty}^{\infty} (z - \mu_Z(t))^2 f_Z(z;t) dz$$

Definitions

The random variables are usually correlated in time (or in space for spatial random functions). This means that the covariance $COV[Z(t_1), Z(t_2)] \neq 0$. Note that the further apart the points (e.g. t_1 and t_2) the lower the covariance. This is estimated as

$$\begin{aligned} COV[Z(t_1), Z(t_2)] &= E[(Z(t_1) - \mu_Z(t_1))(Z(t_2) - \mu_Z(t_2))] \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(z_1 - \mu_Z(t_1))(z_2 - \mu_Z(t_2)) f_Z(z_1, z_2; t_1, t_2) dz_1 dz_2 \end{aligned}$$

if $t_1 = t_2$, $COV[Z(t_1), Z(t_2)] = \sigma_Z^2(t)$.
For n different locations t_i where $i = 1, 2, \cdots, n$, the probability measure that a realization of $Z(t)$ has the values between $z_1 < z(t_1) \leq z_1 + dz_1$, $z_2 < z(t_2) \leq z_2 + dz_2$, $\cdots$, $z_n < z(t_n) \leq z_n + dz_n$. The associated multivariate pdf $f_{Z(t_1),Z(t_2),\cdots,Z(t_n)}(z_1, z_2, \cdots, z_n; t_1, t_2, \cdots, t_n)$

$$f_{Z(t_1),Z(t_2),\cdots,Z(t_n)}(z_1, z_2, \cdots, z_n; t_1, t_2, \cdots, t_n) =$$
$$\lim_{\partial z_1,\cdots,\partial z_n \to 0} \frac{Pr(z_1 < z(t_1) \leq z_1 + \partial z_1, z_2 < z(t_2) \leq z_2 + \partial z_2, \cdots, z_n < z(t_n) \leq z_n + \partial z_n)}{\partial z_1 \partial z_2 \cdots \partial z_n}$$

The multivariate pdf of a random function is sometime referred to as **multipoint pdf**, because it relates to random variables at different points or locations. Note that random function is fully characterized if the multivariate pdf for any set of points is known.

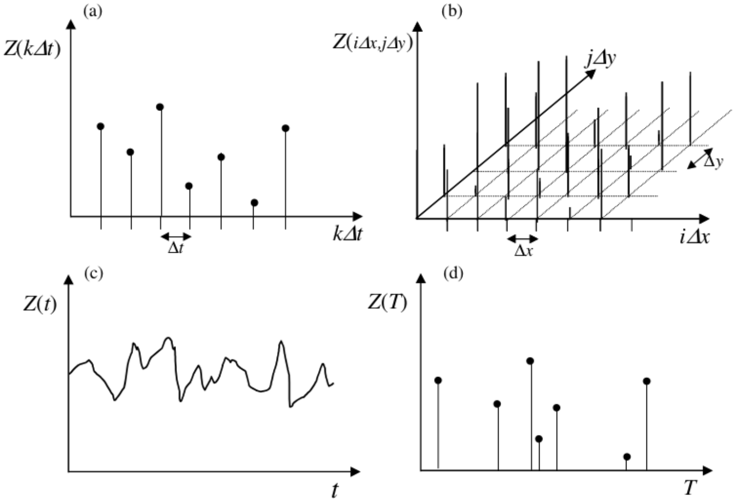
Types of random functions

Random functions can be classified as continuous whether the random variable is continuous (e.g. maximum annual river flows) or discrete whether the random variable is discrete (e.g. number of annual rainy days). Other way to classify a random function is based its domain (see figure). In the case of a continuous random function, this can be defined at:

1. all locations in time, space or space-time: $Z(t)$, $Z(\mathbf{x})$ or $Z(\mathbf{x}, t)$.
2. discrete points in time, space or space-time, where $Z(k\Delta t)$ $(k = 1, 2, \cdots, K)$ is referred to as a **random time series**, and $Z(i\Delta x, j\Delta y)$ $(i = 1, 2, \cdots, I;$ $j = 1, 2, \cdots, J)$ as a **lattice process**.
3. random times or random locations is space and time: $Z(T)$, $Z(\mathbf{X})$ or $Z(\mathbf{X}, T)$. This process is called **compound point process**. Note that the occurrence of the points at random coordinates in space and time is called a **point process**. If the occurrence of such a point is associated with the occurrence of a random variable (e.g. the occurrence of a thunder storm cell at a certain location in space with a random intensity of rainfall) it is called a compound point process.

The same distinction can be made for discrete-valued random functions: $D(t)$, $D(k\Delta t)$ or $D(T)$; $D(\mathbf{x})$, $D(i\Delta x, j\Delta y)$ or $D(\mathbf{X})$.

Types of random functions



This figure shows the different types of random functions based on the way they are defined on the functional domain: a) random time series, b) lattice process, c) continuous-time random function and d) compound point process.

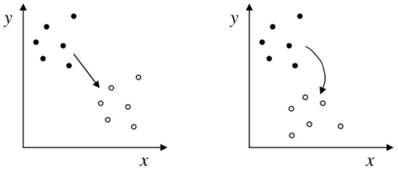
Stationary random functions

Strict stationary random functions

An special type of random functions are the **strict stationary random functions**. A strict stationary random functions isa random function for which its multivariate pdf is invariant under translation. This means that for any set of N locations and for any translations in time Δt, we have

$$f(z_1, z_2, \cdots, z_N; t_1, t_2, \cdots, t_N) = \\ f(z_1, z_2, \cdots, z_N; t_1 + \Delta t, t_2 + \Delta t, \cdots, t_N + \Delta t) \quad \forall t_i, \Delta t$$

This assume that a configuration of points on the time axis can move backward and forward in time and the multivariate pdf does not change. For the spatial domain, strict stationarity means spatial no changes in the pdf if points are translated but no rotated. For instance a strict random function in 2D (see figure) the two sets of locations in the left-hand figure have the same multivariate pdf, while the locations in the right-hand figure necessarily does not. If the multivariate pdf of the random function is invariant under rotation, this is called a **statistically isotropic random function**.



Stationary random functions

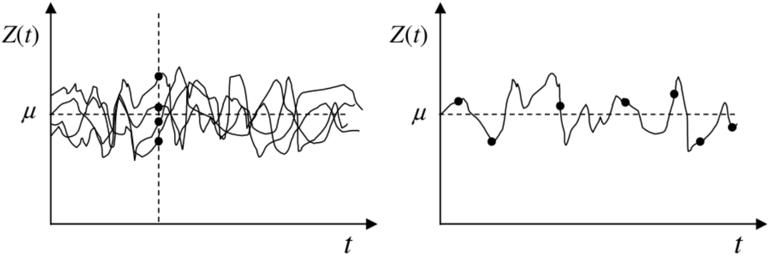
Ergodic random functions

- To estimate the statistical properties of a random function, a large number of realizations must be drew.
- This is the case, for instance, of throwing a dice, from where one can generate multiple realizations.
- In reality this is not possible because we only have one realization: reality. This means that all relevant statistics of the random function are estimated from a single realization.
- In general, **ergodicity** deal with stationarity in random functions.
- To analyse ergodicity, statistical properties such as the mean, the standard deviation and the covariance are examined.
- Stationarity tells that all these properties are invariant no matter where you are (e.g. in time).
- For instance, for multiple realizations at point t_1 of $Z(t)$, the normal procedure to compute $\mu_Z(t_1)$ would be to average these realizations. However, this is impossible because in reality we only have one realization.
- If the random function is stationary the pdf $f_Z(z;t)$ at any location (in time) is the same and thus the statistical properties.
- This also means that within any single realisation we have at every location a sample from the same pdf $f_Z(z;t)$.

Stationary random functions

Ergodic random functions

- So, the mean can also be estimated if we take a sufficient number of samples from a single realisation, such as our reality (see figure below)



- This property of a random function, i.e. being able to estimate statistical properties of a random function from a large number of samples of a single realisation is called **ergodicity**.
- So for a random function is **ergodic** if it is stationary.
- Also for a random function to be ergodic, the samples from the single realisation should be taken from a large enough period of time or, in the spatial case, large enough area.

Ergodic random functions

A formal definition of ergodicity of a random function in its mean

$$\lim_{T\to\infty} \frac{1}{T}\int_T z(t)dt = \int_{-\infty}^{\infty} z f_Z(z;t)dz = \mu_Z$$

where T is an interval of time. This equation means that the integral over the probability distribution (the ensemble) for a given t (term on the left) can be replaced by a temporal (or spatial or spatio-temporal) integral of a very large interval T (or area or volume) (term on the right) to estimate μ_Z. volume).
Similarly, ergodicity of a random function in its mean is defined as

$$\begin{aligned}\lim_{T\to\infty} \frac{1}{T^2}\int_T z(t)z(t+\tau)dt &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(z_1-\mu_Z)(z_2-\mu_Z)f(z_1,z_2;t,t+\tau)dz_1dz_2\\ &= COV[Z(T),Z(t+\tau)] \qquad \forall\tau\end{aligned}$$

where τ is delta time.

Stationary random functions

Second order stationary random functions

A weaker form of stationarity is second order stationarity. This means that the bivariate (or two-point) probability distribution is invariant under translation

$$f(z_1, z_2; t_1, t_2) = f(z_1, z_2; t_1 + \tau, t_2 + \tau) \qquad \forall t_1, t_2, \tau$$